Know your studying self

Choose the option which you think best represents yourself and your behaviour.

Rate based on a scale of 1 to 10

1. I feel that studying is important and necessary

   1 10

2. I study as much as I need to for fulfilling my dreams

   1 10

3. I feel I don't have enough time to study

   1 10

4. I use my time effectively by studying when I need to

   1 10

Studying for Exams

A **workbook** for becoming a **Topper**

Yash Chugh

The Promise I Make to Myself

Make this promise to yourself to stick with the habits that make you who you want to be.

My name is _____
and today, on _____, I pledge to read and do this workbook for at least _____ minutes everyday for _____ (till I become a topper or start studying consistently)

Introduction

Remember what you have learnt till now in your life. If you want, note it down.
Ask yourself, "what do I know? What knowledge do I have?"
You may think, where do I start? I have learnt so much, it would take me too long to remember. And that is precisely what I want you to realise.
Over time, as you keep learning new things each day, from textbooks, teachers and your day-to-day environment, your knowledge keeps increasing.
One example: you're able to read this. That means you know the English language. Maybe you're familiar with another language as well. If so, you know two languages.Then you are bilingual. Read that again. There'll be a test pretty soon. Actually, here it is:
Question: Who can be described as bilingual?
You'll say. It's someone who knows two languages.

Wait. What was that? You may have already learnt this word. But if you had not, you just learnt it!

My point here is : **You have the ability to learn** what you need in class, exams and life.

The Science

Your brain actually changes as you learn and practice and do activities. It forms connections. Between neurons. Kind of like how we seperate waste from useful stuff, your brain decides what's useful based on how long and how frequently you use it.

See a pro football player in action. They are able to control the ball with such skill. They're not born with that skill. They practiced a lot, their brain made the connections stronger, and **they became better than most. And you can too. Let's start**

Chapter 1

Exercise : (needs pencil/pen)

Quickly note down 5 subjects you learn in school, and then write 5 words that come up in your mind when you think of that word.

1.

2.

3

4

5.

Now read it from top to bottom, and try to spot patterns.

Think about this:
Are any of them interrelated?
Did the words you write down first influence your later choice of words?
What significance do they have?
Is there an event or fact you associate with it?
What order did you write the subjects in, and was it easier to write words related to subjects you write first than the last?

This may give you an insight into how your brain works by association. Whenever you come across new stuff, your brain tries to make sense of it. It tries to fit it into your understanding of reality. It can do that by connecting new concepts with what you already know.

Some of these connections are stronger, others are weak. This is where frequent and spaced repetition of a topic comes in. The more it's used, the longer it's stored as a memory. This will be discussed in a later chapter in more detail, in lieu of its importance.

And in this book, we will learn how to make what you need to know have stronger connections, so that you retain it for the long-term, using several scientific strategies. But first, let's see what two classmates, Reshma and Roshni do, when confronted with exams

They are sitting in class, people chattering, when the teacher announces,
“ Class, there will be a test on XYZ topic after two weeks. Please come prepared.”

When they reach home, the thought crosses their mind,
“I need to study for the exam.”
Reshma decides she has enough time and postpones it. She procrastinates until the last day. Roshni doesn't. She uses her time productively by studying intensely with frequent breaks. As she finishes a section, she checkmarks it. The difficulty makes it interesting, but not so hard that it is discouraging. It was like a game.
“It's satisfying and fun to study”, she thinks.

On the day of the exam, Roshni was able to remember everything perfectly. She felt it was too easy. Roshni was the topper. But only this time. Reshma could do it too. She knew it. And she did, after asking Roshni exactly how she did it. And you will know too. (For a quick guide or reference, you can see the Appendix 1 - Roshni's study .)

Before we get started :

I highly recommend you to do the exercises as and when they appear. You will need stationary, a textbook and optionally a phone/laptop if you are interested in technological productivity like free online resources and apps.

Preparation

Depending on the task you have, you can try to get into a calm or energetic mood. Each has its own merits.

1. The peaceful mood.

Your environment affects your mood. But, no matter how peaceful the environment is, if your emotions are chaotic, you won't feel under control and calm. So, let's regulate your emotions.

Exercise : Take deep breaths, and try to be aware of your surroundings. Then, try to pay attention inwards, of your emotions.
Optional: It is immensely beneficial if you try to search for “guided meditation” and do it for 5-10 minutes, especially in the starting.

Some don't feel meditation is useful. It may seem like a waste of time. But trust me, that is only until you haven't felt it yourself. If meditation was an emotion, it would be peace and calmness, like the gentle flowing of a river. And guided meditation can help you get this feeling.

Environment:

Ah! Finally, I got some time to study.

But then a voice calls, “ Beta, halwa khalo “ Or “ “Come eat halwa(a sweet dish) my child” .
‘After that is over, you can give it your all! ‘
No, wait! *Ring ring ring* your friends are calling you to play Valorant and Roblox with them. Will you join them? it's just 30 minutes and they're playing too, you think.
Oh, a new episode of your favourite Netflix show! I'll study after that.
And so it continues. And voila, when the moon comes out and you've got no time left, you wonder, where did it all disappear?

A quiet, secluded place free from distractions can allow you to concentrate

Distractions

Exercise:
Note down distractions you have to tackle nearly everyday. (
Examples :

Phone, laptop, TV, calls, playing, etc.

1.
2.
3.
4.
5.
6.
7.
8.
9.
10.

What exactly happens before you get distracted? What is the cue?
(Example : the notification sound, feeling stressed, getting lost in though, feelingl bored because not challenging enough)

How can you avoid or reduce distractions?
(Example : keeping devices in another room, or switch off/ do not disturb, or meditating. Whatever works for you. Try to remove the event which makes you want the distraction in the first place)

Visualise what you should and would do the next time you get distracted.(It could vary depending on the distraction. It helps to get into the mindset of just not allowing yourself to get distracted. Simply be aware of the distractions from beforehand. Remember the reason, the purpose you were studying. Stay focussed. But if you feel exhausted, you should take a break!)

Now, note it down in a flowchart. (If I get distracted by X,, I will do this or that)

Note down when you get distracted and how.. You will then be able to figure out what steps to take(find the solution here)

Exercise : To reduce cravings or desires for a certain habit.
This has not been scientifically proven, but I think it can work.
Take a distracting habit which you have, and make it boring. Keep doing it in that way.
Example :If watching videos on social media is a bad habit. You like it because it is mostly entertaining. Then think of the most uninteresting boring, obscure and obsolete videos you can think of and search for it. Actively look for it. Then, watch those videos. You will get bored of them - that is why you picked them. Keep doing this in short sessions and the craving will probably reduce.

Quick revision:
So what have you learnt so far
Write down fifteen keywords you can remember and some words related to them. What is your takeaway? Then quickly skim through the book. What did you miss? (This is a studying technique which we will discuss later)

Keywords:
1.
2.
3.
4.
5.
6.
7.
8.
9.
10.
11.
12.
13.
14.
15.

Takeaways:

1.

2.

An aside : The key to Making it Work

If I can summarise the goal of this book in one word, it's consistency. Being consistent in doing homework on time, reviewing notes, studying, and having fun.

You have two main types of memory - short-term memory and long- term memory. In the short term, there is something called a working memory. your brain can only remember upto four packets or bits of information at a time.

Exercise : read the number 05249793. Say it out loud (5+ times)until you think you can say it without looking at it.

Long term memory on the other hand, is retained. But even this can be forgotten if not used or remembered.
It's important to prevent that from happening. After all, what's the point of learning it if you will forget it anyway? It's quite simple, actually.
So, how do we make a short term memory into a long term memory?

We can use a technique called

Spaced Repetition

Exercise: Pick a random weekday(X). And pick a number between 3 to 7(Y).
Now on X day, Y no of weeks ago, what did you learn in school?
Example: On Wednesday, 5 weeks ago, I learnt ..

Chances are high you don't remember, at least not most of it.

But what about something you learnt recently? You remember it more clearly.

So, memory fades over time. But that's not a rule.

We can, in fact, make a past memory even stronger than a recent one. And that's by repeatedly studying it.
Here's a rough timeline you can follow, following a rule I call the 2x rule:

Exercise: Choose something you've to learn. You can try this schedule and plan and see if it works for you.
Timeline:
You learn something in the morning, so you revise in the evening.
Then the next day, two days later, four days later, 8, 16, 32 days later and so on.

But each revision should be different and should employ a variety of techniques which we will discuss.
For now, know that at first, try to understand, and remember. Identify your knowledge gaps and review what you don't remember. Then quiz yourself and teach it to your friends or family. Try to expand the concept, think out of the box, and apply it. One way is to combine the concept with something else you already know to create interesting ideas. Is there a problem you or others face which can be solved partially by using these concepts?

Remember, Studying a topic once can help you understand it. But remember - R for R - repetition for retention

Studying Technique 1-Intense Focus with Breaks

You may recognise this popular technique as Pomodoro, FRILS, HIIT(for exercise), etc.

Studying more isn't always better. You need to be able to retain what you remember. And that is why breaks in between are so important. Scientists think it gives your brain time to process the information. A popular time for sessions is 25-30 minutes then a 5-10 minute break, repeated four times. Then a long 30 minute break. This can help you study for the whole day without feeling burned out - an adequate mix of breaks and studying for optimal efficiency.
Keep in mind, this number isn't fixed for everyone.

Let's apply what we learnt. Whenever you sit down to read this book, remember to check the time.

What's the time now?

Ans:

And then, when you feel like stopping, stop. Take a break, come back later.

Some people prefer 10, while others prefer 60, 90 or even 120 minute sessions. When you feel like you're reading it but stopped understanding it, that could mean you need a break. If you don't, great! You may have entered into the flow state.

Entering FLOW - When you Enjoy Studying AND are Productive

Deep work happens when you are hyperfocussed, so focussed that you find it difficult to stop. This is when you learn more efficiently and have fun. This is what scientists call the Flow state. You are fully absorbed in the activity you are doing, and this can lead to greater satisfaction and achievements.

Exercise: research about the flow state. Many elite stars and athletes are known to use this. Find out a celebrity you know or idolise who uses this.

Try to remember a time when you were so engrossed in an activity that you lost track of time.

Popular people using flow state.

Here are four things

1. The Goldilocks Zone -

    In the Goldilocks and the Threehaven' Goldilocks was able to find the porridge, chair and bed which was *just right*. Similarly, we can make the difficulty of study just right. It may seem at first that the easier the better. But if it's too easy, it becomes boring. Imagine I had you write all the numbers till 1000 10 times. It would be easy, but because of that, it becomes tedious and boring. There's nothing new you're learning.
    What about giving you a questions you need to spend solving. As long as you're making noticeable progress, it's fun. But staying stuck in one place isn't.

    So, we need to adjust the difficulty. It should be challenging, but doable. It should be right at the edge of your current ability.

2. The quiet, distraction- free place
3. Having fun

Techniques to have fun

This is where it gets interesting. Once you start having fun, it's easier to be in flow. And that makes studying even more fun. It's a virtuous cycle. Once you get started in the right direction, you can keep progressing.

There are several techniques to make studying more fun, satisfying and rewarding

1. Meditation before studying:
   Remember how I told you it helps in keeping you calm. Well, it has another use too. There is a scientific basis for this*. Basically, it's more fun relatively when you study after meditating than after watching something stimulating.

   Exercise: (do this regularly)
   If you have permission, watch something you like on your favourite electronic device. But only for 3 minutes. Set a timer, and as soon as it ends, stop watching. Then start meditation (guided meditation if you want). You are thus training your brain to stop associating phones/TV with mindless watching, and enter a state of calmness. It's easier to control

2. Progress bar
   If you like video games, you may know about levelling up and progress bars. It shows how much you have achieved, and that pleasure you get from reaching the next level or rank is a part of the reason why video games are addictive.

Exercise

> Take something you want to learn, and divide it into sections. As you learn each section tick it off.

This is the technique I'm talking about. There is a feeling of satisfaction one gets as you see the progress you make.

Study technique 2 : Active Recall

Reading is a passive form of learning. Active form of learning requires more effort and leads to stronger learning connections.

Exercise : when you have read something, for example this book, try to remember what you're learnt. (We already did this once in previous exercise) .

Now, review what you have learnt.
That means see what you wrote and what you missed.
Have you got the gist of it? What do you need to write?

Whatever you have missed, write it there, preferably with another ink pen.

Then do that again if you would like.

Or use another technique

Study technique 3 - Read topics

Exercise : start from the beginning and read through the headings and subheadings of this book. Then start back here. Check out the exercises and questions too and keywords too.

This process of looking at the main topics can help you get a sense of what you're about to learn. It gives your brain a starting point to build the connections from. As you read through it, you'll then be able to see the connections with the previous and upcoming topics too

Exercise 2 : if you have an old textbook or are able to find it online (for example : cbse.nic.in) , try to read through the headings. You may have forgotten most of it. But as you skim through it, recall. See the diagrams, the equations and questions. Haven't you learnt some of that recently again? What do you remember? By the time you're done, you're surprised how much you've forgotten, but know you remembered a lot too.

Now, recall again what you remember here:

…

Note: This is why metaphors are such an effective way to help you learn. You build new connections based on connections you already have. Analogies are like the threads of a spiders web, they catch the new ideas and slowly the web expands.

Study Technique 4 - Interleaving

Exercise : what have you learnt till now which you can apply ?

We have learnt

1. Frequent and intense sessions
2. Spaced Repetition
3. Meditate for a peaceful mood and quiet, distraction-free environment
4. Make it challenging and interesting.
5. Reading through the headings and using analogies
6. Interleaving

Interleaving is where you take several concepts you have learnt and learn them one by one. But it's also when exam tests have questions from multiple different chapters. Each time you have to switch from a different set of neural connections.
Here, we will use it to combine into an effective study method

Exercise: pick a topic you want to study and use the techniques as you do so.

Study Technique 5 - Develop yourself

Exercise - list 15 daily habits you do which are good for you

1.
2.
3.
4.
5.
6.
7.
8.
9
10.
11.
12.
13.
14.
15.

Exercise - list some habits which detoriarare your health

1.
2.
3.
4.
5.
6.
7.
8.

Exercise-What steps will you take ?

One way to help you do good habits is by making it easier to do so.
Often, we don't even realise because something is just a little difficult we choose to take the easy route. So make these the easiest there is!

If you want to play sports, keep your sports gear and equipment right next to you
If you want to study more, make a plan and decide what you'll study. Then keep the book outt so that when you have time you can get straight to studying.
This is what we used for removing distractions too

Exercise: modify the things in your house.
Take a look at the list you've made and see how you can improve it

Exercise:
Make a pledge to yourself. If all of them seem too daunting, set specific goals and do 2-3 at a time. Some improvement is better than none

I make a pledge to myself _____(name)to do_____ (activity)on _____(date) for at least _____ days

Checklist
Each day: put one tickmark

I make a pledge to myself _____(name)to do_____ (activity)on _____(date) for at least _____ days

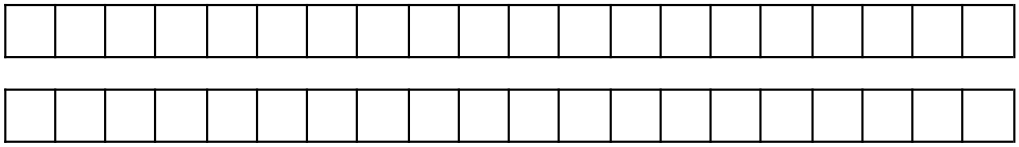
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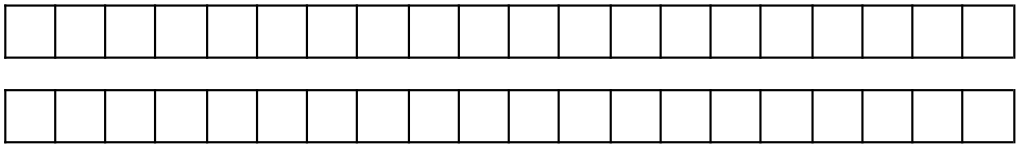
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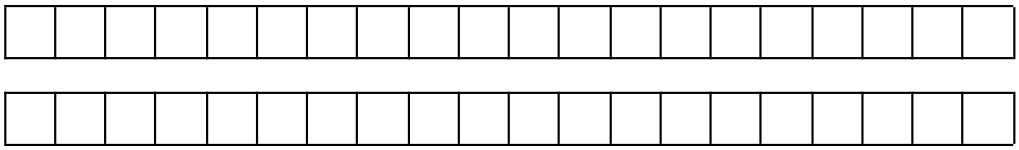
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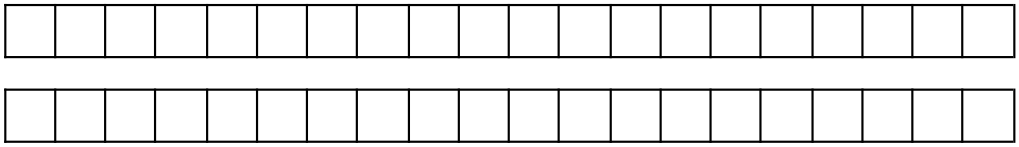
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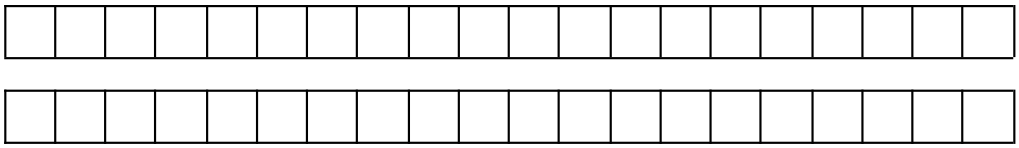
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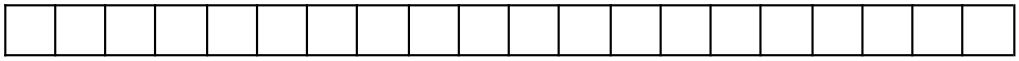
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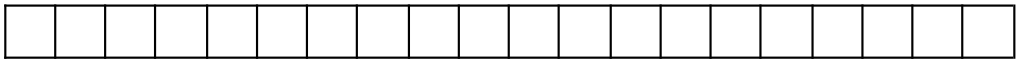
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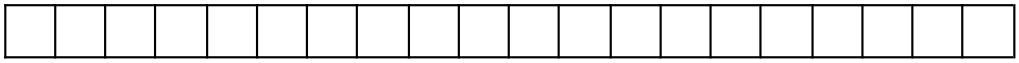




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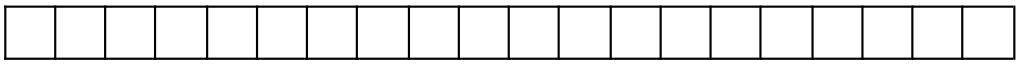
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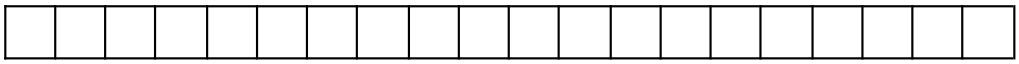




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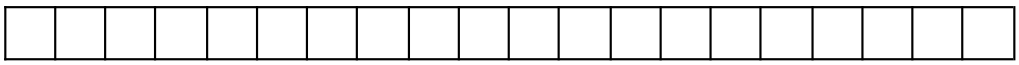
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Checklist
Each day: put one tickmark



Conclusion

Thank you for reading this book. If you've enjoyed it and become a topper, I've fulfilled this purpose

Further reading :

1. Atomic Habits by James Clear
2. Master your Emotions by Thibaut Meurisse

Free Resources:

Khan Academy

YouTube

Brilliant

Quizzes

Kahoot

Pomodoro apps

UCLA Berkeley guided meditations

Ted-ed talk on flow is on Youtube

Sources : Atomic Habits,
YouTube -Ted-ed,
Coursera - Barbara Oakley
The Internet